

Reference excerpt 02-038

Source: arXiv:1604.05177v1 | License: CC BY 4.0 | Word window: 1405-1664

of CSA and ECG trace are shown. The x and y are not correctly detected. Correct waves are indicated. The P, R and T events of the ECG are indicated. 3 DISCUSSION This note describes a methodology to extrapolate some interesting quantitative parameters from the IJV CSA trace, which is an analogue of the JVP. There are two important aspects that must be considered: i) the contouring of the IJV on each sonogram of the video-clip must be as accurate as possible. This means that the algorithm presented in [Sisini et al.(2015)] must be improved to achieve an accuracy close to 100 %, while the current one is about 95 %. ii) The wave detection algorithm must be improved. In fact it is not always sufficient to identify the waves, while they can be identified by the analysis of tracks performed by a human expert. This paper has not discussed the clinical use of these parameters. ACKNOWLEDGEMENT The author wants to thank Giacomo Gadda, Valentina Tavoni and Valentina Sisini to have kindly reviewed the manuscript. REFERENCES [Applefeld(1990)] Applefeld MM. The Jugular Venous Pressure and Pulse Contour. In: Source Clinical Methods: The History, Physical, and Laboratory Examinations. Boston: Butterworths, 1990: Chapter 19. [Beulen et al.(2011)] Beulen, B. et al. Toward noninvasive blood pressure assessment in arteries by using ultrasound. Ultrasound in Medicine, 2011;37(5), 788-797. [Beulen et al.(2010)] Beulen, B., Verkaik, A. C., Bijlens, N., Rutten, M., Van De Vosse, F. (2010). Perpendicular ultrasound velocity measurement by 2D cross correlation of RF